



Day 10: Persistent Storage with Docker Volumes

[General Interview Questions]

1. What is a Docker volume?

Answer:

A Docker volume is a storage mechanism managed by Docker that allows data to persist independently of the container lifecycle.

2. Why do we use Docker volumes?

Answer:

To persist data, share data between containers, and decouple storage from container lifespans.

3. How is a volume different from the container filesystem?

Answer:

Container filesystems are ephemeral and lost when the container is deleted, while volumes persist even after container removal.

4. Where are Docker volumes stored on the host?

Answer:

On Linux systems, Docker volumes are stored in:

`/var/lib/docker/volumes/`

5. What is the command to create a Docker volume?

Answer:

`docker volume create my_volume`

6. How do you list all Docker volumes on your system?

Answer:

docker volume ls

7. How do you inspect a Docker volume?

Answer:

docker volume inspect my_volume

8. What is the command to delete a specific volume?

Answer:

docker volume rm my_volume

9. How can you remove all unused volumes?

Answer:

docker volume prune

10. How do you attach a volume to a container using -v?

Answer:

docker run -v my_volume:/app/data nginx

11. What is the --mount option used for in Docker?

Answer:

It is an alternative to -v that offers a clearer and more structured syntax for attaching volumes or mounts.

12. What is the difference between a named volume and an anonymous volume?

Answer:

- **Named Volume:** Created with a specific name and can be reused.
- **Anonymous Volume:** Created automatically by Docker without a name, harder to manage.

13. What happens to a volume if the container is deleted?

Answer:

The volume remains on the host and can be reused unless explicitly removed.

14. Can multiple containers use the same volume?

Answer:

Yes, multiple containers can read/write to the same volume for data sharing.

15. How do you attach a volume using --mount syntax?

Answer:

```
docker run --mount source=my_volume,target=/app/data nginx
```

16. When would you use a volume instead of a bind mount?

Answer:

When you want Docker-managed, portable, and isolated persistent storage.

17. What are some common use cases for Docker volumes?

Answer:

- Persisting database data
- Sharing config files between containers
- Storing logs and output files

18. Can you back up and restore Docker volumes?

Answer:

Yes, using docker run with tar and mounting the volume to back up or restore data.

19. What does docker volume prune do?

Answer:

It removes all unused (dangling) volumes from the host.

20. Is it possible to specify volume options during creation?

Answer:

Yes, options like driver or labels can be specified:

```
docker volume create --driver local --label team=data my_volume
```